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Elasticity & Modulus of Materials

Stress-Strain Dynamics, Young's Modulus, Shear & Bulk Moduli, Poisson's Ratio, Elastic Potential Energy Density, and Coupled Thermal-Mechanical Stresses

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Part I: Foundations of Elasticity & The Stress-Strain Relationship

Chapter 1: Microscopic & Macroscopic Mechanics of Deformable Solids

In classical rigid-body mechanics, physical bodies are idealized as perfectly undeformable entities where the distance between any two constituent particles remains immutable under the application of arbitrary forces. While this abstraction is exceptionally productive for analyzing gross orbital trajectories, center-of-mass kinematics, and static equilibrium, it collapses when confronted with real matter. All real engineering solids — from structural steels and aerospace titaniums to biological bones and elastomer seals — deform when subjected to mechanical loading.

1.1 Microscopic Atomic Origins of Elastic Restoring Forces

At the microscopic level, the resistance of a crystalline or amorphous solid to shape distortion originates in the fundamental electrostatic and quantum-mechanical interactions between neighboring atoms or ions. The potential energy $U(r)$ of an adjacent atomic pair separated by distance r represents a competition between long-range attractive forces (Coulombic, van der Waals, or metallic valence attractions) and short-range Pauli exclusion repulsion of overlapping electron shells:

$$U(r) = -\frac{A}{r^m} + \frac{B}{r^n} \qquad (n > m)$$

At the equilibrium separation distance $r = r_0$, the attractive and repulsive forces exactly balance, meaning the net interatomic force $F(r)$ vanishes:

$$F(r_0) = -\left.\frac{dU}{dr}\right|_{r=r_0} = 0$$

For infinitesimal macroscopic displacements, the atoms undergo minute perturbations $x = r - r_0$ about the bottom of the potential well. Performing a Taylor series expansion of $U(r)$ about r_0 yields:

$$U(r) \approx U(r_0) + \left.\frac{dU}{dr}\right|_{r_0}(r - r_0) + \frac{1}{2}\left.\frac{d^2U}{dr^2}\right|_{r_0}(r - r_0)^2 = U(r_0) + \frac{1}{2}k_{atomic}(r - r_0)^2$$

Because the first derivative vanishes at equilibrium, the bottom of the potential well is parabolic. Differentiating with respect to displacement gives a linear restoring force:

$$F_{restoring} = -\frac{dU}{dr} = -k_{atomic}(r - r_0)$$

This establishes the microscopic foundation of **Hooke's Law**: macroscopic linear elasticity is the direct consequence of harmonic oscillations within the parabolic minimum of interatomic potential energy wells.

> **◎ Physical Insight: The Modulus Measures Atomic Bond Stiffness**
>
> The macroscopic elastic modulus of a material is not an arbitrary empirical fitting parameter; it is directly proportional to the curvature of the interatomic potential well at its minimum: $Y \sim k_{atomic} / r_0$. Materials with deep, narrow potential wells (such as diamond and tungsten) possess exceptionally high interatomic spring constants and consequently enormous Young's moduli, whereas materials with shallow, broad potential wells (such as lead or polymers) deform with far lower applied stress.

1.2 Macroscopic Definition of Stress: Engineering vs. True Stress

To formulate a continuum theory independent of test specimen geometry, external loads are normalized by the cross-sectional area across which they act. **Stress** (σ) represents the intensity of internal reaction forces distributed over an infinitesimal internal plane:

$$\sigma = \lim_{\Delta A \to 0} \frac{\Delta F_n}{\Delta A}$$

In engineering practice, two distinct definitions of normal stress are distinguished:

- **Engineering (Nominal) Stress (σ_0 or σ):** The applied tensile or compressive load F divided by the *original, undeformed* cross-sectional area A_0:

$$\sigma = \frac{F}{A_0} \qquad [\mathrm{N/m^2 = Pa}]$$

- **True Stress (σ_{true}):** The instantaneous load F divided by the *instantaneous actual* cross-sectional area A at that load level:

$$\sigma_{true} = \frac{F}{A} = \frac{F}{A_0} \cdot \frac{A_0}{A} = \sigma\,(1 + \varepsilon)$$

(valid under plastic deformation where volume remains strictly conserved, $A\,L = A_0\,L_0$).

1.3 Tensile vs. Compressive Stress

Normal stresses act perpendicular (normal) to the cross-sectional plane of interest:

- **Tensile Stress (σ > 0):** The internal forces act directed away from the cut surface, stretching the material longitudinally. Failure occurs via ductile necking and void coalescence or brittle cleavage tearing.
- **Compressive Stress (σ < 0):** The internal forces act directed toward the surface, shortening the material along the loading axis. While isotropic polycrystalline metals exhibit virtually identical Young's moduli in tension and compression, compressive failure in slender members is almost universally governed by **geometric instability (Euler buckling)** rather than intrinsic material yield strength:

$$P_{cr} = \frac{\pi^2\,Y\,I}{(K\,L)^2}$$

In brittle materials (concrete, masonry, bone, rock), compressive strength is typically 5 to 10 times higher than tensile strength because compressive loads squeeze microcracks shut, preventing rapid mode-I crack propagation.

1.4 Macroscopic Definition of Strain: Engineering vs. True Strain

Strain is a dimensionless measure of normalized geometric displacement:

- **Engineering (Cauchy) Strain (ε):** The change in total length ΔL divided by the original gage length L_0:

$$\varepsilon = \frac{\Delta L}{L_0} = \frac{L - L_0}{L_0}$$

- **True (Logarithmic / Hencky) Strain (ε_{true}):** The continuous integration of incremental length changes over instantaneous length:

$$\varepsilon_{true} = \int_{L_0}^{L} \frac{dL}{L} = \ln \frac{L}{L_0} = \ln(1 + \varepsilon)$$

◆ Proof of Equivalence for Infinitesimal Elastic Deformations

Using the Taylor expansion of the natural logarithm $\ln(1 + x) = x - x^2/2 + x^3/3 - \ldots$:

$$\varepsilon_{true} = \ln(1 + \varepsilon) = \varepsilon - \frac{1}{2}\varepsilon^2 + O(\varepsilon^3)$$

In engineering elasticity, strains rarely exceed $\varepsilon = 0.002$ (0.2%). For $\varepsilon = 0.002$, the quadratic term is $\varepsilon^2/2 = 0.000002$ (0.1% relative error). Hence, for all linear elastic calculations, engineering strain and true strain are mathematically identical: $\varepsilon_{true} \approx \varepsilon$.

Chapter 2: The Engineering Stress-Strain Curve & Material Regimes

When a standardized tensile specimen (such as an ASTM round bar) is pulled at a controlled quasi-static strain rate, the recorded relationship between nominal stress σ and nominal strain ε reveals the complete mechanical fingerprint of the material.

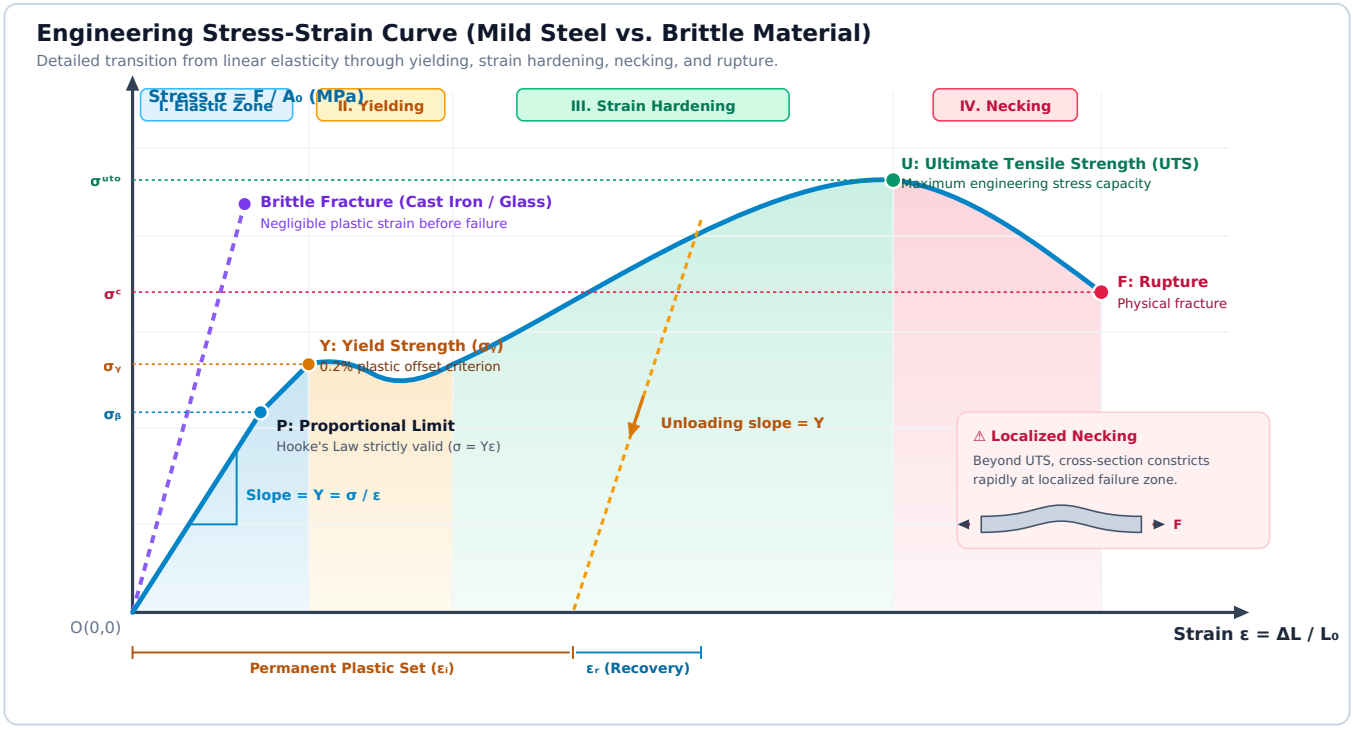


Figure 5.1: Complete annotated engineering stress-strain curve for a ductile alloy (mild steel) compared against a brittle material, detailing the proportional limit, yield strength, UTS, necking instability, and permanent plastic recovery set.

2.1 The Four Characteristic Deformation Regimes

As illustrated in Figure 5.1, the ductile stress-strain trajectory traverses four distinct physical regimes:

1. **Region I: Linear Elastic Regime (0 → P → Y):** From the origin to the **Proportional Limit (P)**, stress is strictly linear with strain ($\sigma = Y\,\varepsilon$). If unloaded anywhere within the elastic zone, the specimen retracts along the exact same path to zero strain without permanent distortion. At the **Yield Point (Y)**, the maximum stress σ_y is reached beyond which irreversible atomic slip begins. For materials without a sharp yield drop, the standard **0.2% Offset Yield Strength** ($\sigma_{0.2\%}$) is defined by constructing a line parallel to the elastic slope starting at $\varepsilon = 0.002$.

2. **Region II: Yielding Plateau & Upper/Lower Yielding (Y → Plateau):** In low-carbon structural steels, a sharp drop occurs from an Upper Yield Point to a Lower Yield Point. This behavior is governed by the pinning of dislocations by interstitial carbon and nitrogen solute atoms (**Cottrell atmospheres**). Once the applied stress unpins these dislocations, they propagate at lower stress, forming visible shear bands across the gage length (**Lüders bands**).
3. **Region III: Strain Hardening / Work Hardening (Plateau → U):** Beyond the yield plateau, further plastic deformation requires steadily increasing stress. As dislocations glide along crystallographic slip planes, they intersect, cross-slip, and form dense dislocation tangles and cell walls. These internal defects act as barriers to further dislocation motion, raising the macroscopic shear resistance. The cross-sectional area contracts uniformly along the entire gage length.
4. **Region IV: Ultimate Tensile Strength & Localized Necking (U → F):** The peak of the engineering curve represents the **Ultimate Tensile Strength (UTS, σ_{uts})**. At point U, the geometric reduction in cross-sectional area outpaces the rate of material strain hardening:

$$\frac{d\sigma_{true}}{d\varepsilon_{true}} = \sigma_{true} \qquad \text{(Considère's Instability Criterion)}$$

Plastic deformation abruptly localizes into a neck, dropping nominal stress until final **Rupture (F)** at σ_f.

2.2 Unloading, Plastic Recovery & Permanent Set

If a structural member is strained into the plastic regime (point *A* beyond yield with total strain ε_{tot}) and subsequently unloaded, the unloading trajectory follows an elastic recovery line whose slope matches the original Young's modulus *Y*:

$$\varepsilon_{total} = \varepsilon_p + \varepsilon_e = \varepsilon_p + \frac{\sigma}{Y}$$

At $\sigma = 0$, an irreversible plastic strain ε_p remains (the **permanent set**). Subsequent reloading retraces this recovery path back up to point *A*, establishing a higher effective yield strength $\sigma_{y,new} \approx \sigma_A$ (the principle of industrial **cold working** or **strain hardening**).

2.3 Ductile vs. Brittle Material Behavior

- **Ductile Materials (Structural steel, aluminum, copper, titanium):** Exhibit extensive plastic deformation before fracture (ε_f typically 15%–45%), noticeable necking, and a characteristic 45° "cup-and-cone" shear failure surface providing visual warning prior to structural collapse.
- **Brittle Materials (Gray cast iron, glass, ceramics, concrete):** Deform elastically up to sudden cleavage rupture with negligible plastic flow ($\varepsilon_f < 1\%$) perpendicular to maximum tensile stress planes.

2.4 Energy Storage: Modulus of Resilience vs. Modulus of Toughness

- **Modulus of Resilience (U_r):** Elastic strain energy stored per unit volume without permanent set:

$$U_r = \int_0^{\varepsilon_y} \sigma \, d\varepsilon = \frac{1}{2}\,\sigma_y\,\varepsilon_y = \frac{\sigma_y^{\,2}}{2\,Y} \qquad [\mathrm{J/m^3}]$$

High resilience demands high yield strength σ_y and low Young's modulus *Y* (e.g. spring steels).

- **Modulus of Toughness (U_t):** Total strain energy absorbed per unit volume up to rupture: $U_t = \int_0^{\varepsilon_f} \sigma \, d\varepsilon$ [J/m^3]. Demands balancing high strength with ductility (brittle ceramics have high σ_y but very low toughness).

Part II: Fundamental Elastic Moduli & Multiaxial Response

Chapter 3: Young's Modulus & Hooke's Law in Uniaxial Tension/Compression

Under uniaxial tensile or compressive loading, the ratio of normal stress to normal strain within the proportional limit defines the primary measure of solid tensile stiffness: **Young's Modulus (Y or E)**, named after the British polymath Thomas Young (1773–1829).

3.1 Mathematical Formulation of Hooke's Law

In its classic one-dimensional form, Hooke's Law states that stress is directly proportional to strain:

$$\sigma = Y\,\varepsilon \quad \Leftrightarrow \quad \frac{F}{A} = Y\,\frac{\Delta L}{L_0}$$

Solving for the absolute axial elongation ΔL of a prismatic bar of uniform initial cross-sectional area A and original length L_0 under an axial tensile force F gives:

$$\Delta L = \frac{F\,L_0}{A\,Y} \qquad [\text{m}]$$

Because strain ε is dimensionless, the SI unit of Young's Modulus is identical to stress: **Pascals (Pa = N/m^2)**. For structural metals, values are typically reported in **Gigapascals (GPa = 10^9 Pa)**: structural steel (200 GPa), titanium alloy (114 GPa), aluminum 6061 (70 GPa), and cortical bone (18 GPa).

3.2 The Equivalent Spring Constant Mapping: k = YA / L₀

Rewriting Hooke's Law reveals an immediate structural isomorphism with an ideal linear spring:

$$F = Y\,A\,\frac{\Delta L}{L_0} = k_{eq}\,\Delta L \quad \Rightarrow \quad k_{eq} = \frac{Y\,A}{L_0} \qquad [\text{N/m}]$$

This mapping allows any system of elastic structural rods, trusses, or columns to be analyzed using standard electrical circuit / spring network formulas:

- **Bars in Series (Co-axial segments carrying the same force F):**

$$\Delta L_{total} = \Delta L_1 + \Delta L_2 \quad \Rightarrow \quad \frac{1}{k_{eq}} = \frac{1}{k_1} + \frac{1}{k_2} = \frac{L_1}{Y_1\,A_1} + \frac{L_2}{Y_2\,A_2}$$

- **Bars in Parallel (Side-by-side elements sharing the same extension ΔL):**

$$F_{total} = F_1 + F_2 \quad \Rightarrow \quad k_{eq} = k_1 + k_2 = Y_1\,A_1\frac{}{L_0} + Y_2\,A_2\frac{}{L_0}$$

3.3 Elongation of Non-Uniform Bars & Self-Weight Integration

When the cross-sectional area $A(x)$ or the internal axial force $F(x)$ varies continuously along the longitudinal coordinate x (from fixed support $x = 0$ to tip $x = L$), Hooke's Law is applied differential element by element:

$$d(\Delta L) = \varepsilon(x)\,dx = \frac{F(x)}{A(x)\,Y}\,dx \quad \Rightarrow \quad \Delta L = \int_0^L \frac{F(x)}{A(x)\,Y}\,dx$$

◆ Rigorous Proof: Exact Elongation of a Heavy Hanging Cable Under Self-Weight

Consider a vertical cable of uniform area A, length L, mass density ρ, and Young's modulus Y, suspended from a rigid ceiling under its own gravitational weight, carrying a concentrated bottom payload M_{load}.

At distance x below the ceiling support, the cross-section must support the payload plus the mass of cable hanging below x:

$$F(x) = M_{load}\,g + \rho\,A\,g\,(L - x)$$

The total elongation is found by integrating along the entire cable length from $x = 0$ to $x = L$:

$$\Delta L = \int_0^L \frac{M_{load}\,g + \rho\,A\,g\,(L - x)}{A\,Y}\,dx = \frac{M_{load}\,g\,L}{A\,Y} + \frac{\rho\,g}{Y}\int_0^L (L - x)\,dx$$

Evaluating the linear integral $\int_0^L (L - x)\,dx = [L\,x - x^2/2]_0^L = L^2/2$:

$$\Delta L_{self} = \frac{\rho\,g\,L^2}{2\,Y} = \frac{1}{2}\,\frac{(\rho\,A\,L)\,g\,L}{A\,Y} = \frac{1}{2}\,\frac{M_{cable}\,g\,L}{A\,Y}$$

Conclusion: The self-weight elongation is *exactly half* of what would occur if the entire cable weight were concentrated at the lower tip! The factor of 1/2 arises because internal tension ramps linearly from zero at the bottom to $M_{cable}g$ at the top.

Chapter 4: Shear Modulus, Shear Stress & Torsional Mechanics

While normal stress acts perpendicular to an internal cross-section causing longitudinal stretch or compression, **shear stress** acts tangentially along the surface plane, sliding internal atomic layers parallel to one another.

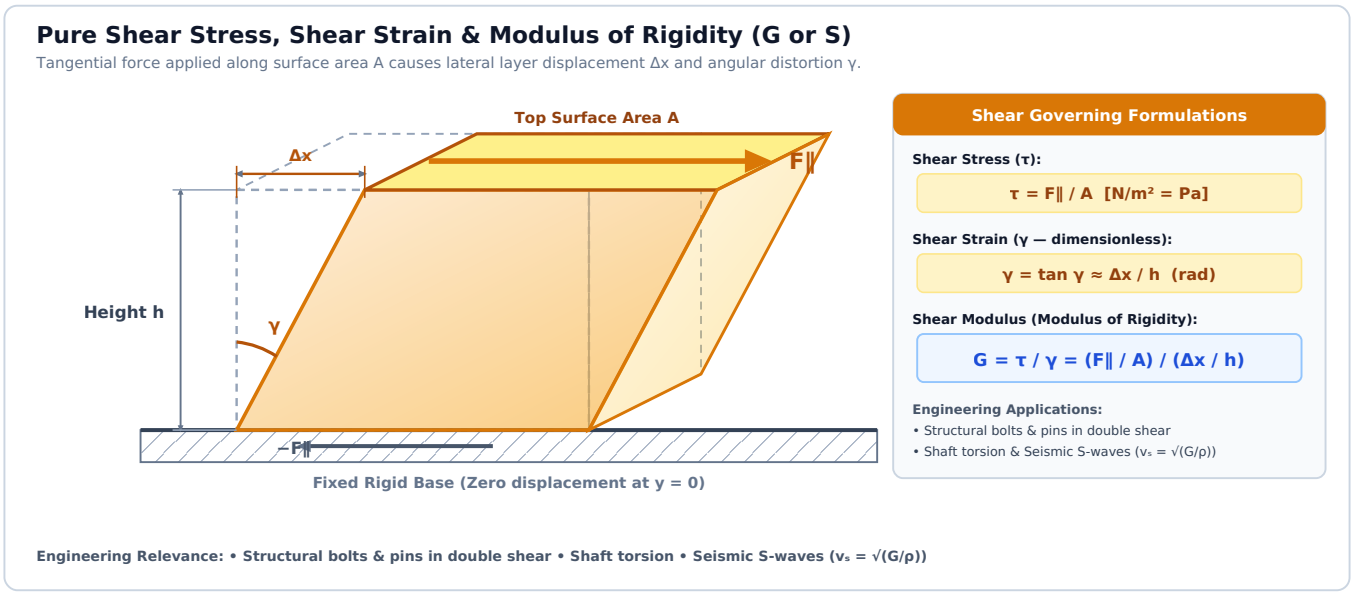


Figure 5.2: Pure shear deformation of a solid block subjected to tangential force F∥ over surface area A, illustrating horizontal shift Δx, shear angle γ, and complementary base reaction.

4.1 Governing Formulations: Shear Stress (τ) and Shear Strain (γ)

Referring to Figure 5.2, consider a rectangular block of height h and top surface area A firmly anchored to a rigid base:

- **Shear Stress (τ):** The tangential shear force $F_{\parallel}$ divided by the surface area A:

$$\tau = \frac{F_{\parallel}}{A} \qquad [\text{Pa} = \text{N/m}^2]$$

- **Shear Strain (γ):** The horizontal displacement Δx of the top plane relative to the fixed base, divided by the vertical separation height h:

$$\gamma = \tan\gamma \approx \frac{\Delta x}{h} \qquad [\text{radians — dimensionless}]$$

For small elastic deformations ($\gamma \ll 1$ rad), the small-angle approximation $\tan\gamma \approx \gamma$ holds with supreme precision.

4.2 Shear Modulus (Modulus of Rigidity: G or S)

Within the proportional limit, shear stress is strictly proportional to shear strain:

$$\tau = G\,\gamma \quad \Rightarrow \quad G = \frac{\tau}{\gamma} = \frac{F_{\parallel}/A}{\Delta x / h} = \frac{F_{\parallel}\,h}{A\,\Delta x} \qquad [\text{Pa}]$$

The shear modulus G (frequently denoted S in introductory texts) measures the resistance of a material to angular distortion and internal sliding. For common engineering metals, G is roughly 35% to 40% of Young's modulus Y (e.g., steel: $Y \approx 200$ GPa, $G \approx 80$ GPa; aluminum: $Y \approx 70$ GPa, $G \approx 26$ GPa).

4.3 Complementary Shear Stresses & Shaft Torsion

Static moment equilibrium of an infinitesimal cubic stress element requires that shear stress cannot exist on a single plane in isolation. Taking rotational equilibrium about the center of an element ($\sum M_z = 0$) proves that:

$$\tau_{xy} = \tau_{yx} \qquad \text{(Reciprocity of Shear Stresses)}$$

Every shear stress τ on a horizontal plane induces an identical, complementary shear stress τ on vertical orthogonal planes. When a circular shaft of length L and outer diameter d is subjected to twisting torque T, the internal shear stress varies linearly with radius r, reaching a maximum at the outer surface:

$$\tau_{\max} = \frac{T\,(d/2)}{J} = \frac{16\,T}{\pi\,d^3} \qquad \text{and} \qquad \theta = \frac{T\,L}{J\,G}$$

where $J = \pi\,d^4 / 32$ is the polar moment of inertia of the circular cross-section, and θ is the total angle of twist in radians.

4.4 Geophysics & Seismic Waves: Proof that the Outer Core is Liquid

The physical distinction between normal and shear stresses plays a paramount role in global geophysics. In a continuous medium, mechanical disturbances propagate as two distinct wave types:

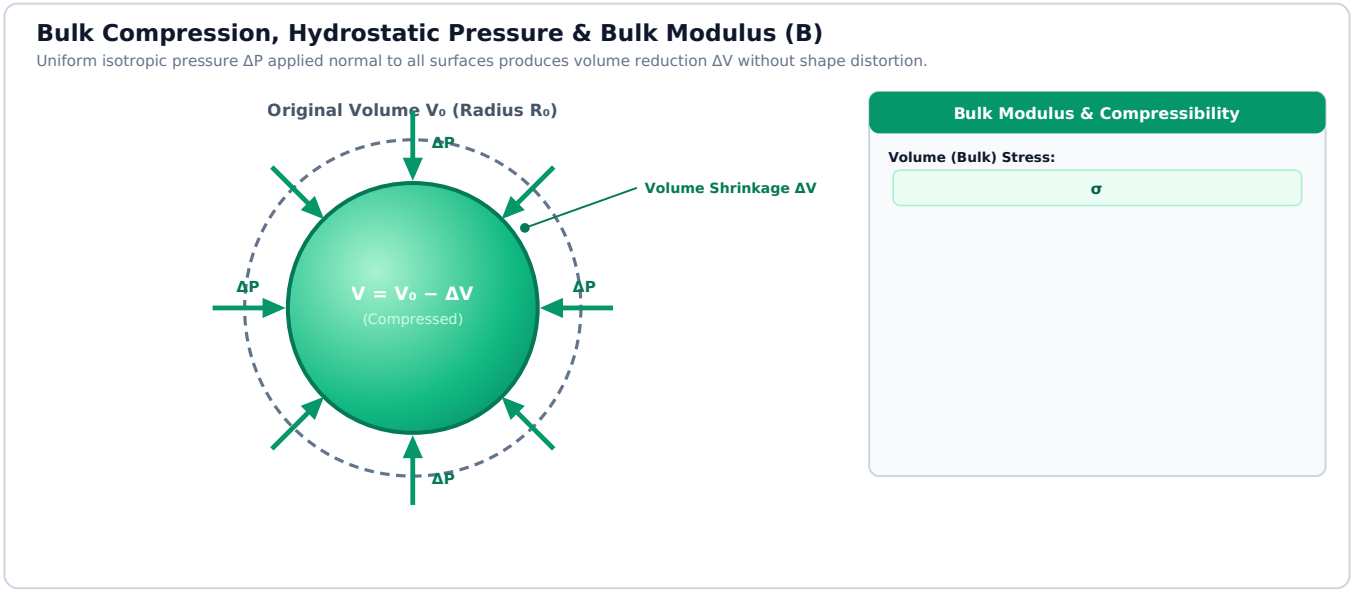
- **Primary Longitudinal Pressure Waves (P-waves):** Compressional waves propagating at wavespeed $v_p = \sqrt{((K + 4G/3)/\rho)}$.
- **Secondary Transverse Shear Waves (S-waves):** Pure shear waves propagating at wavespeed:

$$v_S = \sqrt{G\rho}$$

Because fluids (both liquids and gases) possess zero static shear rigidity ($G_{fluid} = 0$), transverse shear stresses cannot be sustained. Consequently, **S-waves cannot propagate through fluids**. In 1906, Richard Dixon Oldham observed that earthquake S-waves completely vanish across a large "shadow zone" on the opposite side of Earth, providing indisputable physical proof that Earth possesses a molten liquid outer core!

Chapter 5: Bulk Modulus, Hydrostatic Pressure & Compressibility

When an object is immersed in a pressurized fluid (such as a submarine diving in deep ocean trenches or an alloy specimen inside a hydraulic pressure chamber), external forces act perpendicular to all boundary surfaces simultaneously.



$_V = \Delta P = F\perp / A$ [Pa = N/m²] Volume (Bulk) Strain: $\varepsilon_V = \Delta V / V_0$ (< 0 under compression) Bulk Modulus (B — Always Positive): $B = -\Delta P / (\Delta V / V_0) = -V_0 (dP / dV)$ Compressibility (k or β): $k = 1 / B$ [$Pa^{-1} = m^2/N$] • Water: B = 2.2 GPa (nearly incompressible) Physical Benchmark: • Water: B = 2.2 GPa • Steel: B = 160 GPa • Air (isothermal): $B = P_0 \approx 101$ kPa

Figure 5.3: Hydrostatic bulk compression of a solid body under uniform fluid pressure increment ΔP, producing fractional volume shrinkage ΔV / V₀ with conserved geometric shape.

5.1 Hydrostatic Stress & Volumetric Strain

Referring to Figure 5.3:

- **Bulk (Volume) Stress (ΔP):** The uniform increase in normal hydrostatic pressure applied over all faces:

$$\Delta P = \frac{F_\perp}{A} \quad [\text{Pa} = \text{N/m}^2]$$

- **Bulk (Volume) Strain (ε_V):** The fractional reduction in volume:

$$\varepsilon_V = \frac{\Delta V}{V_0} = \frac{V - V_0}{V_0} \quad [\text{dimensionless, } < 0 \text{ under compression}]$$

5.2 Bulk Modulus (B or K) and Compressibility (k or β)

The **Bulk Modulus (B)** is defined as the negative ratio of hydrostatic pressure change to volumetric strain:

$$B = -\frac{\Delta P}{\Delta V / V_0} = -V_0 \frac{dP}{dV} \quad [\text{Pa}]$$

The explicit negative sign ensures that B is a positive scalar quantity, since an increase in hydrostatic pressure ($\Delta P > 0$) invariably produces a decrease in volume ($\Delta V < 0$).

The reciprocal of the bulk modulus is the **Compressibility** (k or β):

$$k = \beta = \frac{1}{B} = -\frac{1}{V_0}\frac{dV}{dP} \quad [\text{Pa}^{-1} = \text{m}^2/\text{N}]$$

Liquids and solids exhibit high bulk moduli and extremely low compressibility (e.g. water: $B \approx 2.2$ GPa, $k \approx 4.5 \times 10^{-10}$ Pa^{-1}; steel: $B \approx 160$ GPa). Gases, in contrast, have bulk moduli proportional to their ambient pressure P.

5.3 Bulk Modulus of an Ideal Gas: Isothermal vs. Adiabatic

◆ Derivation of Ideal Gas Bulk Moduli and the Speed of Sound in Air

1. Isothermal Compression (Constant Temperature, T = const):
From the ideal gas law, $P\,V = n\,R\,T = const$. Differentiating gives:

$$P\,dV + V\,dP = 0 \quad \Rightarrow \quad \frac{dP}{dV} = -\frac{P}{V} \quad \Rightarrow \quad B_T = -V\frac{dP}{dV} = P$$

The isothermal bulk modulus of an ideal gas equals its absolute pressure P!

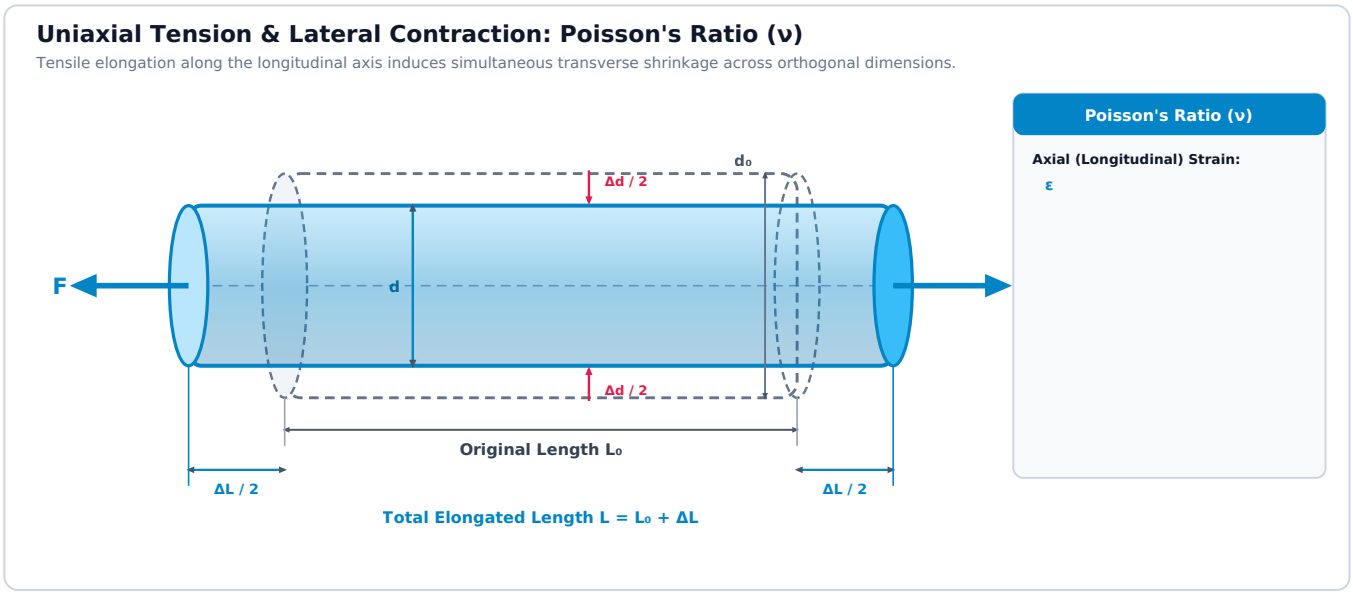
2. Adiabatic Compression (Rapid oscillations, zero heat transfer, Q = 0):
For rapid acoustic compressions, the process obeys $P\,V^{\gamma} = const$, where $\gamma = C_p/C_v$ is the adiabatic heat capacity ratio (for air, $\gamma \approx 1.40$):

$$V^{\gamma}\,dP + \gamma\,P\,V^{\gamma-1}\,dV = 0 \quad \Rightarrow \quad \frac{dP}{dV} = -\gamma\frac{P}{V} \quad \Rightarrow \quad B_{ad} = \gamma\,P$$

Isaac Newton originally estimated the speed of sound using isothermal elasticity $v = \sqrt{(P/\rho)}$, finding ~280 m/s (~16% too low). Pierre-Simon Laplace corrected this by recognizing sound waves are adiabatic: $v = \sqrt{(\gamma\,P\,/\,\rho)} \approx 343$ *m/s* at 20°C.

Chapter 6: Poisson's Ratio & Multiaxial Volumetric Coupling

When an elastic solid is stretched along one axis, the atomic bonds along that axis are elongated, causing the transverse atomic spacings to contract to minimize energy. This lateral contraction is governed by **Poisson's Ratio (ν)**, formulated by Siméon Denis Poisson (1781–1840).



long = ΔL / L₀ > 0 Lateral (Transverse) Strain: $\varepsilon_{trans} = \Delta d / d_0 < 0$ Fundamental Definition: $\nu = -\varepsilon_{trans} / \varepsilon_{long}$ Physical Limits: • Cork: ν ≈ 0.00 (no lateral bulge) • Steels: ν ≈ 0.28 – 0.33 • Rubber: ν ≈ 0.499 (ΔV ≈ 0) Physical Limits: • Cork: ν ≈ 0.00 • Steels: ν ≈ 0.28 – 0.33 • Rubber: ν ≈ 0.499 (ΔV ≈ 0) • Auxetic: −1.0 ≤ ν < 0

Figure 5.4: Uniaxial tensile elongation along longitudinal axis accompanied by transverse shrinkage across the diameter, illustrating the definition and sign convention of Poisson's ratio ν.

6.1 Definition & Sign Convention

Referring to Figure 5.4, if a cylindrical bar of original diameter d_0 and original length L_0 is loaded by uniaxial tensile stress $\sigma_x = F/A$:

- **Axial (Longitudinal) Strain:** $\varepsilon_{axial} = \Delta L / L_0 > 0$
- **Lateral (Transverse) Strain:** $\varepsilon_{transverse} = \Delta d / d_0 < 0$

Poisson's ratio is defined as the negative ratio of transverse strain to axial strain:

$$\nu = -\frac{\varepsilon_{transverse}}{\varepsilon_{axial}} = -\frac{\Delta d / d_0}{\Delta L / L_0} \quad \text{[dimensionless]}$$

The explicit negative sign ensures that for all ordinary materials that thin out when stretched, ν is a positive number.

6.2 Fractional Volume Change under Uniaxial Tension

◆ **Derivation: Volumetric Strain and the Incompressibility Limit ν = 0.5**

Consider a rectangular block of initial dimensions L_x, L_y, L_z and original volume $V_0 = L_x L_y L_z$ subjected to uniaxial tensile strain ε along the x-axis.
By Poisson's effect, the principal strains are: $\varepsilon_x = \varepsilon$, $\varepsilon_y = -\nu\varepsilon$, $\varepsilon_z = -\nu\varepsilon$.
The deformed volume is:

$$V = L_x(1 + \varepsilon) \cdot L_y(1 - \nu\varepsilon) \cdot L_z(1 - \nu\varepsilon) = V_0 (1 + \varepsilon)(1 - \nu\varepsilon)^2$$

Expanding and retaining only first-order linear terms in strain ($\varepsilon^2 \approx 0$):

$$V \approx V_0 [1 + \varepsilon - 2\nu\varepsilon + O(\varepsilon^2)] = V_0 [1 + \varepsilon (1 - 2\nu)]$$

The fractional volume change (volumetric strain) is therefore:

$$\frac{\Delta V}{V_0} = \frac{V - V_0}{V_0} = \varepsilon\,(1 - 2\nu)$$

Critical Engineering Conclusions:
1. **Conventional Solids (ν < 0.5):** Under tension (ε > 0), $1 - 2\nu > 0$, so volume *increases* ($\Delta V > 0$).
2. **Incompressible Materials (ν = 0.5):** If ν = 0.5, then $1 - 2(0.5) = 0$, meaning $\Delta V \equiv 0$ identically! Vulcanized natural rubber (ν ≈ 0.499) and soft biological tissues maintain essentially constant volume during deformation.
3. **Zero-Poisson Materials (ν = 0):** For bottle cork (ν ≈ 0), $1 - 2(0) = 1$, so $\Delta V/V = \varepsilon$. When a cork is driven into a wine bottle, it does not expand laterally, avoiding damaging the bottle neck!

6.3 3D Generalized Hooke's Law & Thermodynamic Bounds

In a general 3D triaxial stress state (σ_x, σ_y, σ_z), the total strain along each axis is obtained by superposing the direct axial strain and the Poisson transverse contractions from the other two perpendicular stresses:

$$\varepsilon_x = \frac{1}{Y}\left[\,\sigma_x - \nu\,(\sigma_y + \sigma_z)\,\right]$$

$$\varepsilon_y = \frac{1}{Y}\left[\,\sigma_y - \nu\,(\sigma_z + \sigma_x)\,\right]$$

$$\varepsilon_z = \frac{1}{Y}\left[\,\sigma_z - \nu\,(\sigma_x + \sigma_y)\,\right]$$

Summing all three orthogonal strains gives the total volumetric strain:

$$\frac{\Delta V}{V_0} = \varepsilon_x + \varepsilon_y + \varepsilon_z = \frac{1 - 2\nu}{Y}\,(\sigma_x + \sigma_y + \sigma_z)$$

For thermodynamic stability of an isotropic continuum, the strain energy function must be strictly positive definite under any arbitrary loading, requiring both $G > 0$ and $B > 0$. From the relations $Y = 2G(1 + \nu)$ and $Y = 3B(1 - 2\nu)$, this imposes the fundamental mathematical bounds:

$$-1.0 \le \nu \le +0.5 \qquad \text{(Thermodynamic Stability Limits)}$$

Materials with negative Poisson's ratio ($-1 \le \nu < 0$) are known as **auxetic materials**. When pulled longitudinally, their re-entrant cellular microstructure unfolds, causing them to *expand* laterally. Auxetic metamaterials are utilized in bulletproof vests, high-performance running shoes, and expandable medical stents.

Chapter 7: Interrelations Among Isotropic Elastic Moduli

A central theorem of classical continuum mechanics states that for any **isotropic and homogeneous linear elastic medium**, all elastic behavior is completely determined by exactly **two independent material constants**. All other moduli — Young's modulus Y, shear modulus G, bulk modulus B, Poisson's ratio ν, and Lamé's first parameter λ — are algebraic functions of any chosen pair.

7.1 Fundamental Conversion Equations

▶ The Master Moduli Interrelation Equations

1. Young's & Shear Modulus Link: $Y = 2\,G\,(1 + \nu) \quad \Leftrightarrow \quad G = \dfrac{Y}{2(1 + \nu)}$

2. Young's & Bulk Modulus Link: $Y = 3\,B\,(1 - 2\nu) \quad \Leftrightarrow \quad B = \dfrac{Y}{3(1 - 2\nu)}$

3. Poisson's Ratio from G & B: $\nu = \dfrac{3B - 2G}{2(3B + G)} = \dfrac{3B - 2G}{6B + 2G}$

4. Young's Modulus from G & B: $Y = \dfrac{9\,B\,G}{3B + G}$

5. Bulk Modulus from Y & G: $B = \dfrac{G\,Y}{3(3G - Y)}$

6. Shear Modulus from Y & B: $G = \dfrac{3\,B\,Y}{9B - Y}$

◎ Physical Interpretation of Moduli Consistency

Notice that as a material approaches incompressibility ($\nu \to 0.5$), the denominator of Bulk Modulus $3(1 - 2\nu) \to 0$, causing $B \to \infty$. At the same time, $Y = 2G(1 + 0.5) = 3G$. For an incompressible elastomer (such as rubber), Young's modulus is exactly three times its shear modulus ($Y = 3G$), while its resistance to hydrostatic pressure is infinite!

Part III: Energy Methods, Dynamic Loading & Thermal Coupling

Chapter 8: Elastic Potential Energy Density & Dynamic Impact Dynamics

When external forces perform work on a deformable body within its elastic limit, the work is stored internally as recoverable **Elastic Potential Energy** (strain energy). Upon release of the load, this internal strain energy is returned to the surroundings, powering spring mechanisms, recoiling archery bows, and driving mechanical vibrations.

8.1 Work Done During Deformation & Strain Energy

Consider a prismatic bar stretched quasi-statically by an axial load increasing gradually from 0 to F. Because the restoring force scales linearly with displacement according to Hooke's Law ($F(x) = k\,x$), the total work done by the external force is:

$$W = \int_0^{\Delta L} F(x)\,dx = \int_0^{\Delta L} k\,x\,dx = \frac{1}{2}\,k\,(\Delta L)^2 = \frac{1}{2}\,F\,\Delta L \qquad [\text{J} = \text{N}\cdot\text{m}]$$

The factor of 1/2 is fundamental: it represents the triangular area beneath the linear load-displacement diagram. Expressing ΔL in terms of stress and strain ($\Delta L = \varepsilon\,L_0$ and $F = \sigma\,A$) gives the total stored strain energy U:

$$U = \frac{1}{2}\,(\sigma\,A)\,(\varepsilon\,L_0) = \frac{1}{2}\,\sigma\,\varepsilon\,(A\,L_0) = \frac{1}{2}\,\sigma\,\varepsilon\,V_0 = \frac{F^2\,L_0}{2\,A\,Y}$$

8.2 Elastic Strain Energy Density (u)

Dividing total strain energy by the specimen volume $V_0 = A\,L_0$ yields the **Strain Energy Density (u)**, representing energy stored per unit volume:

$$u = \frac{U}{V_0} = \int_0^{\varepsilon} \sigma\,d\varepsilon = \frac{1}{2}\,\sigma\,\varepsilon = \frac{1}{2}\,Y\,\varepsilon^2 = \frac{\sigma^2}{2\,Y} \qquad [\text{J/m}^3 = \text{Pa}]$$

Analogous expressions govern shear and volumetric deformations:

- **Pure Shear Strain Energy Density:**

$$u_{shear} = \frac{1}{2}\,\tau\,\gamma = \frac{1}{2}\,G\,\gamma^2 = \frac{\tau^2}{2\,G} \qquad [\text{J/m}^3]$$

  For a circular shaft under torque T: $U_{torsion} = T^2\,L\,/\,(2\,J\,G)$.

- **Hydrostatic Bulk Strain Energy Density:**

$$u_{bulk} = \frac{1}{2}\,\Delta P\,\varepsilon_v = \frac{1}{2}\,B\,\frac{\Delta V}{V_0}^2 = \frac{(\Delta P)^2}{2\,B} \qquad [\text{J/m}^3]$$

8.3 Dynamic Impact Loading & The Dynamic Amplification Factor

In many engineering scenarios — such as an elevator falling onto an arresting buffer or an anchor dropped onto a crane cable — loads are applied dynamically with kinetic energy rather than quasi-statically.

◆ Derivation of the Dynamic Amplification Factor and Maximum Impact Deflection

A mass m is released from height h above a collar attached to the lower tip of a vertical rod of stiffness $k = Y A / L_0$. Let Δ_{max} be the maximum instantaneous downward displacement.

By conservation of energy, the gravitational potential energy lost by the falling mass equals the peak elastic strain energy stored in the stretched rod:

$$m\,g\,(h + \Delta_{max}) = \frac{1}{2}\,k\,\Delta_{max}^{2} \quad \Rightarrow \quad \Delta_{max}^{2} - 2\,\Delta_{st}\,\Delta_{max} - 2\,\Delta_{st}\,h = 0$$

where $\Delta_{st} = m\,g / k = W\,L_0 / (A\,Y)$ is the static deflection under the hanging load. Solving the quadratic equation for the positive root yields the peak dynamic elongation Δ_{max} and the **Dynamic Amplification Factor (n)**:

$$\Delta_{max} = \Delta_{st}\left[\,1 + \sqrt{1 + \frac{2\,h}{\Delta_{st}}}\,\right] \quad \Rightarrow \quad \mathbf{n} = \frac{\Delta_{max}}{\Delta_{st}} = \frac{\sigma_{max}}{\sigma_{st}} = 1 + \sqrt{1 + \frac{2\,h}{\Delta_{st}}}$$

Two Fundamental Limiting Regimes:

• **Suddenly Applied Load (h = 0):** $n = 1 + \sqrt{1 + 0} = \mathbf{2.0}$. An instantaneously released load generates *exactly twice* the stress and deflection of a static load!

• **Drop from Large Height ($h \gg \Delta_{st}$):** $n \approx \sqrt{(2h / \Delta_{st})}$. Because $\Delta_{st} \propto 1/k$, a stiffer bar has a smaller static deflection, which paradoxically produces a *vastly higher* dynamic magnification and peak stress upon impact!

⚠ The Rigidity Trap in Impact Absorbers

Designers intuitively assume that making a structural member thicker and stiffer will make it safer against impact. However, because impact absorbs a fixed amount of kinetic energy $E = U = \sigma^2 V / (2Y)$, increasing stiffness reduces deformation distance, forcing peak impact force and dynamic stress to skyrocket! Effective impact absorbers (crumple zones, climbing ropes, aircraft landing gears) are designed to be compliant and ductile, maximizing energy dissipation over a large deformation stroke.

Chapter 9: Coupled Thermal Stress & Constrained Expansion

When the temperature of a solid changes, thermal agitation shifts the mean equilibrium spacing of atoms within their asymmetric potential wells. If the solid is free to expand, it changes length without experiencing internal stress. However, if thermal expansion is prevented by rigid supports, massive internal stresses are generated.

9.1 Superposition of Thermal & Elastic Strains

For a temperature change $\Delta T = T - T_0$, the unrestrained free thermal strain is proportional to the linear coefficient of thermal expansion α:

$$\varepsilon_{thermal} = \alpha\,\Delta T \quad [K^{-1} \text{ or } {}^{\circ}C^{-1}]$$

If external mechanical forces or rigid boundary constraints simultaneously exert a normal stress σ, the total strain is obtained by linear superposition (within the elastic range):

$$\varepsilon_{total} = \varepsilon_{thermal} + \varepsilon_{elastic} = \alpha\,\Delta T + \frac{\sigma}{Y}$$

Multiplying by the original length L_0 gives the total net deformation:

$$\Delta L_{total} = \alpha\, L_0\, \Delta T + \frac{F\, L_0}{A\, Y}$$

9.2 Rigidly Constrained Boundaries (ΔL = 0)

Consider a bar of length L_0 placed snugly between two unyielding, immovable walls. Because the total length cannot change, the kinematic compatibility boundary condition is strictly:

$$\Delta L_{total} = 0 \quad \Rightarrow \quad \varepsilon_{total} = 0$$

Substituting into the superposition relation:

$$\alpha\,\Delta T + \frac{\sigma_{thermal}}{Y} = 0 \quad \Rightarrow \quad \sigma_{thermal} = -\,Y\,\alpha\,\Delta T \quad [\text{Pa}]$$

The corresponding internal reaction force exerted against the restraining walls is:

$$F_{thermal} = |\sigma_{thermal}|\, A = Y\, A\, \alpha\, \Delta T \quad [\text{N}]$$

Fundamental Physical Observations:

1. **Length Independence:** The magnitude of thermal stress $\sigma_{th} = Y\,\alpha\,\Delta T$ is **entirely independent of the length L_0 of the bar**! A 1-meter bar and a 100-kilometer continuous railway rail experience identical thermal stress under the same temperature shift.
2. **Sign Convention:**
   - **Heating (ΔT > 0):** The bar attempts to expand, so the rigid walls push back inward, inducing **compressive thermal stress ($\sigma_{th} < 0$)**. If the compressive stress exceeds the Euler buckling threshold, the bar will buckle catastrophically sideways.
   - **Cooling (ΔT < 0):** The bar attempts to contract, so the walls pull outward, inducing **tensile thermal stress ($\sigma_{th} > 0$)**. In brittle materials (concrete roadways, ceramic coatings), cooling induces tensile cracking and joint fracture.

9.3 Constrained Expansion with Initial Clearance Gap (δ)

In civil structures (such as bridge finger expansion joints and railway expansion gaps), an intentional gap δ is provided between structural sections.

- **Case 1: Free Expansion ($\alpha L_0 \Delta T \le \delta$):** The thermal expansion does not close the gap. The bar expands freely by $\Delta L = \alpha\, L_0\, \Delta T$, and internal thermal stress remains strictly zero: $\sigma_{thermal} = 0$.
- **Case 2: Constrained Expansion ($\alpha L_0 \Delta T > \delta$):** The expanding bar bridges the gap and pushes against the opposing support. The total allowable elongation is limited to δ:

  $$\Delta L_{total} = \delta \quad \Rightarrow \quad \alpha\, L_0\, \Delta T + \frac{\sigma\, L_0}{Y} = \delta$$

  Solving for the induced compressive stress:

  $$\sigma_{thermal} = -\,Y\,\left[\alpha\, L_0\, \Delta T - \delta L_0\right] = -\,Y\,\alpha\,\Delta T\,\left[L_0 - \delta / (\alpha\,\Delta T) L_0\right]$$

Part IV: Hand Calculation Shortcuts (⚡ Fast-Tracks)

In professional engineering assessments and university physics competitions, solving elasticity problems from scratch by setting up full differential equations or multi-variable systems is time-consuming and error-prone. The following five **Fast-Track Hand Calculation Shortcuts** distill key continuum mechanics relationships into rapid mental-math rules, dimensional scaling laws, and physical invariants.

TRICK 1 Equivalent Spring Constant Mapping ($k = YA/L_0$) for Composite Assemblies — 10 Seconds

Standard Method (Slow): Writing separate Hooke's Law equations $\Delta L_1 = F_1 L_1 / (A_1 Y_1)$ and $\Delta L_2 = F_2 L_2 / (A_2 Y_2)$, solving for displacements, and eliminating internal forces through multi-variable substitution (3–5 minutes).

⚡ **Fast Hand Calculation Shortcut:** Treat each bar segment as a discrete spring with stiffness $\mathbf{k = Y A / L_0}$. Immediately apply standard circuit rules:

- **Series (End-to-End):** $1/k_{eq} = \sum (L_i / (Y_i A_i)) \Rightarrow \Delta L = F / k_{eq}$.
- **Parallel (Co-axial):** $k_{eq} = \sum k_i$. Load carried by member 1 is simply the stiffness fraction: $\mathbf{F_1 = F_{total}\,[k_1 / \sum k_i]}$.

Example: Steel (k_s = 400 MN/m) and aluminum (k_a = 280 MN/m) in parallel: steel instantly carries 400/(400+280) = **58.8%** of total load!

TRICK 2 Rapid Geometric Scaling Ratios & Self-Weight Average Factor — 5 Seconds

Standard Method (Slow): Recalculating cross-sectional areas $A = \pi d^2/4$ from scratch, plugging in numbers with 10^{-6} prefixes, and performing long decimal division.

⚡ **Fast Hand Calculation Shortcut:** Recognize that axial elongation scales as $\mathbf{\Delta L \propto F \cdot L / d^2}$.

- **Doubling diameter (d → 2d):** Cuts stretch by a factor of **4** ($\Delta L' = \Delta L / 4$).
- **Doubling length (L → 2L):** Doubles stretch ($\Delta L' = 2\,\Delta L$) | **Doubling both:** Halves stretch ($\Delta L' = \Delta L / 2$).
- **Self-Weight Rule of Half:** Elongation of a uniform suspended cable under its own weight is **exactly 1/2** of an equivalent tip point load: $\mathbf{\Delta L_{self} = 0.5 \cdot (M_{cable}\, g\, L) / (A\, Y)}$.

TRICK 3 Fractional Volume Change Shortcut via Poisson's Ratio ($\Delta V/V \approx \varepsilon(1 - 2\nu)$) — 3 Seconds

Standard Method (Slow): Calculating elongated length $L = L_0(1 + \varepsilon)$, contracted radius $r = r_0(1 - \nu\varepsilon)$, expanding volume $\pi r^2 L$, and subtracting V_0.

⚡ **Fast Hand Calculation Shortcut:** Directly use the first-order volumetric strain identity: $\mathbf{\Delta V / V_0 = \varepsilon \cdot (1 - 2\nu)}$.

- **Structural Steel ($\nu \approx 0.30$):** $\Delta V / V_0 = \mathbf{0.40\,\varepsilon}$ | **Aluminum ($\nu \approx 0.33$):** $\Delta V / V_0 \approx \mathbf{(1/3)\,\varepsilon}$
- **Elastomers / Rubber ($\nu = 0.50$):** $1 - 2(0.50) = 0 \Rightarrow \mathbf{\Delta V / V_0 \equiv 0}$ (strictly volume-conserving / incompressible).

Example: A steel bar stretched by $\varepsilon = 0.1\%$ increases in volume by exactly $0.4 \times 0.1\% = \mathbf{0.04\%}$ without touching a calculator!

TRICK 4 Ocean Hydrostatic Compression & Seismic Wavespeed Link (v = √(B/ρ)) 5 Seconds

Standard Method (Slow): Computing hydrostatic pressure from P = ρ g h with multi-digit densities, converting to Pascals, and evaluating bulk modulus ratios.

⚡ **Fast Hand Calculation Shortcut:** Use the universal ocean depth conversion rule: **every 100 meters of seawater depth adds ≈ 1.0 MPa (10 atmospheres) of pressure**.
• Depth h = 1,000 m ⇒ ΔP ≈ 10 MPa = 0.010 GPa.
• Deepest trench h = 11,000 m ⇒ ΔP ≈ 110 MPa = 0.11 GPa.
• Because water has bulk modulus B ≈ 2.2 GPa, the fractional volume shrinkage of seawater is: **$|\Delta V / V_0| = \Delta P / B \approx 0.010 / 2.2 \approx 0.0045 \approx 0.45\%$ per kilometer of depth**.
• **Fluid Wavespeed Shortcut:** Speed of sound in any fluid is **$v = \sqrt{(B / \rho)}$**. For water: $\sqrt{(2.2 \times 10^9 / 1000)} = \sqrt{(2.2 \times 10^6)} \approx$ **1480 m/s**.

TRICK 5 Thermal Stress Constrained Bar Quick-Check ($\sigma^{\text{th}} = -Y \alpha \Delta T$) 5 Seconds

Standard Method (Slow): Setting up thermal expansion $\Delta L_{th} = \alpha L \Delta T$, mechanical compression $\Delta L_{mech} = -F L / (A Y)$, equating to zero, and solving for F and σ.

⚡ **Fast Hand Calculation Shortcut:** Memorize the intrinsic thermal stress coefficient **$S_{th} = Y \cdot \alpha$** [MPa/°C]:
• **For Structural Steel:** Y = 200 GPa = 200×10^3 MPa, α = 12×10^{-6} /°C.
⇒ **$S_{th} = 200 \times 10^3 \times 12 \times 10^{-6} = 2.40$ MPa / °C**.
Every 1°C temperature change generates exactly **2.4 MPa of stress**!
A summer temperature rise of ΔT = +50°C immediately induces **$\sigma = -2.4 \times 50 = -120$ MPa of compressive stress**.
• **For Aluminum Alloy:** Y = 70 GPa, α = 24×10^{-6} /°C ⇒ **$S_{th} = 70 \times 0.024 = 1.68$ MPa / °C**.

★ Master Fast-Track Summary Matrix

Shortcut Domain	Governing Relationship	Standard Formula	Mental-Math Rule
Spring Stiffness	Series / Parallel Rods	$k = Y A / L_0$	Parallel: $k_{eq} = \sum k_i$ • Series: $1/k_{eq} = \sum 1/k_i$
Geometric Scaling	Diameter & Length	$\Delta L \propto F L / d^2$	Double diameter ⇒ ¼ stretch • Self-weight ⇒ 0.5 factor
Volume Change	Poisson Coupling	$\Delta V / V_0 \approx \varepsilon(1 - 2\nu)$	Steel: 0.40 ε • Aluminum: 0.34 ε • Rubber: 0
Ocean Depth	Bulk Compression	$\Delta P = \rho g h$ • $\Delta V/V = \Delta P/B$	~1.0 MPa per 100 m • ~0.45% water shrinkage / km
Thermal Stress	Constrained Bar	$\sigma_{th} = -Y \alpha \Delta T$	Steel: 2.40 MPa / °C • Aluminum: 1.68 MPa / °C

Part V: Step-by-Step Worked Problems with Physical Sanity Checks

In professional structural engineering and university physics competitions, formulating an exact quantitative solution demands more than simply plugging numbers into isolated formulas. Real-world structural mechanics requires a rigorous, five-phase analytical methodology that establishes coordinate systems, ensures kinematic displacement compatibility, applies linear constitutive laws, carries explicit SI units through algebraic evaluations, and subjects the final result to asymptotic boundary sanity checks.

▶ The 5-Phase Continuum Mechanics Problem-Solving Protocol

1. **Phase 1 — Problem Statement & Free-Body Isolation:** Identify the primary loading modes (uniaxial tension, pure shear, hydrostatic pressure, dynamic impact, or thermal restraint) and construct free-body equilibrium cuts.
2. **Phase 2 — Given Parameters & Unit Normalization:** Tabulate all mechanical moduli (Y, G, B, ν), geometries, and yield criteria strictly in fundamental SI base units (m, Pa, N, kg, s).
3. **Phase 3 — Symbolic Closed-Form Derivation:** Solve differential balance and compatibility equations symbolically prior to inserting numerical values, making governing scaling laws explicit.
4. **Phase 4 — Numerical Calculation with Explicit Units:** Compute precise intermediate values (masses, stiffnesses, strains) and evaluate engineering Factors of Safety against material failure thresholds.
5. **Phase 5 — Physical Sanity & Boundary Limit Check:** Test asymptotic extremes ($\Delta T \to 0$, $h \to 0$, $L \to 0$, $\nu \to 0.5$) and verify energy conservation or independent Fast-Track shortcuts.

◆ Master Worked Problems Directory & Coverage Roadmap

Problem	Physical System & Loading	Governing Law	Core Engineering Takeaway
Problem 5.1	Vertical Mining Cable with Self-Weight	$\Delta L = (W_{load} + \frac{1}{2} W_{cable}) L / (A Y)$	Exact 0.5 self-weight stretch factor integration
Problem 5.2	Bridge Clevis Joint in Double Shear	$\tau = F / (2 A_{pin})$, $\Delta x = \tau h / G$	Double shear halves pin shear stress and doubles safety
Problem 5.3	Mariana Trench Spherical Hull	$\Delta V/V = -P / B$, $\sigma = P R / (2 t)$	Hadal-depth 110 MPa pressure & 0.10% contraction
Problem 5.4	Bimetallic Column under Thermal Shift	$\sigma_{th} = -Y \alpha \Delta T$, $F = \sum Y_i A_i \alpha_i \Delta T$	Rigid constraint thermal stress is length-independent
Problem 5.5	Crane Arrestor Drop Impact	$n = 1 + \sqrt{(1 + 2h / \Delta_{st})}$	150-fold dynamic impact force & rigidity trap

Worked Problem 5.1: Suspended Vertical Heavy Mining Cable with End Load & Self-Weight

1. Problem Statement:

A deep vertical mine shaft utilizes a high-strength structural steel cable of length L = 500 m and nominal diameter d = 24 mm to suspend an ore cage payload of mass M_{load} = 3,000 kg. The steel has mass density ρ = 7,850 kg/m^3, Young's modulus Y = 200 GPa, and yield strength σ_y = 600 MPa. Taking gravitational acceleration g = 9.80 m/s^2:

(a) Determine the mass of the steel cable itself and the total gravitational load at the top support.
(b) Calculate the maximum tensile stress occurring at the topmost cable cross-section, and evaluate the factor of safety against yielding.
(c) Derive the total elongation ΔL of the cable accounting for both the payload and the cable's distributed self-weight.
(d) Determine the percentage of total elongation contributed by the cable's own weight.

2. Given Parameters & Knowns:

- Cable: Length L = 500 m, diameter d = 24 mm ⇒ Area $A = \pi d^2 / 4 \approx 4.5239 \times 10^{-4}$ m^2
- Ore cage payload: M_{load} = 3,000 kg ⇒ $W_{load} = M_{load} g = 3000 \times 9.80 = 29{,}400$ N
- Material: Density ρ = 7,850 kg/m^3, Y = 200 GPa = 2.0×10^{11} Pa, σ_y = 600 MPa = 6.0×10^8 Pa | g = 9.80 m/s^2

3. Symbolic Algebraic Derivation:

1. Total suspended weight and peak tensile stress occurring at the top support (x = 0):

$$W_{total} = W_{load} + W_{cable} = (M_{load} + \rho A L) g \quad \Rightarrow \quad \boldsymbol{\sigma_{max}} = \frac{W_{total}}{A} = \frac{M_{load} g}{A} + \rho g L$$

2. At depth x below the top support, internal force is $F(x) = M_{load} g + \rho A g (L - x)$. Integrating local strain gives total elongation:

$$\Delta L = \int_0^L \frac{F(x)}{A Y}\, dx = \frac{W_{load} L}{A Y} + \frac{\rho g}{Y} \int_0^L (L - x)\, dx = \frac{W_{load} L}{A Y} + \frac{1}{2} \frac{W_{cable} L}{A Y}$$

4. Numerical Calculation with Explicit SI Units:

- **Cable Mass & Weight:** $M_{cable} = \rho A L = 7850 \times (4.5239 \times 10^{-4}) \times 500 =$ **1,775.6 kg** | $W_{cable} = 1775.6 \times 9.80 =$ **17.40 kN**
- **Total Top Force & Stress:** $W_{total} = 29.40 + 17.40 =$ **46.80 kN** | $\sigma_{max} = 46{,}801 / (4.5239 \times 10^{-4}) =$ **103.45 MPa**
- **Factor of Safety:** $N = \sigma_y / \sigma_{max} = 600 / 103.45 =$ **5.80** (safely exceeds deep-mine safety threshold $N \geq 5.0$)
- **Payload Elongation:** $\Delta L_{load} = (29{,}400 \times 500) / [(4.5239 \times 10^{-4}) \times (2.0 \times 10^{11})] =$ **162.5 mm** (0.1625 m)
- **Self-Weight Elongation:** $\Delta L_{self} = 0.5 \times (17{,}401 \times 500) / (9.0478 \times 10^7) =$ **48.1 mm** (0.0481 m)
- **Total Stretch & Fraction:** $\Delta L = 162.5 + 48.1 =$ **210.6 mm (0.211 m)** | **Self-Weight Fraction:** 48.1 / 210.6 = **22.8%**

✓ **5. Physical Sanity & Boundary Check:**

- **Boundary Consistency:** As payload mass $M_{load} \to 0$, ΔL smoothly approaches ΔL_{self} = 48.1 mm. As cable length $L \to 0$, $\Delta L \to 0$.
- **Strain Magnitude:** Peak strain $\varepsilon_{max} = 103.45 \times 10^6 / (200 \times 10^9) = 0.052\% \ll$ yield strain $\varepsilon_y = 0.30\%$ (strictly linear elastic).
- **Self-Weight Rule of Half:** Even though cable mass is 59% of payload mass, its stretch contribution is only 48.1 / 162.5 = 29.6% due to the 0.5 integration factor!

Worked Problem 5.2: Structural Riveted Lap Joint under Double Shear

1. Problem Statement:

A heavy structural clevis joint connects a bridge tie-rod carrying an axial tensile force F = 45 kN. The joint is secured by a single hardened alloy steel shear pin in **double shear**. The pin has diameter d = 16 mm, shear modulus G = 80 GPa, and ultimate shear strength τ_{uts} = 320 MPa. The clearance span across the shear gap between plates is h = 25 mm.

(a) Calculate the cross-sectional shear area of each shear plane and the total resisting area.
(b) Determine the average shear stress τ acting across the pin.
(c) Compute the factor of safety N against ultimate shear failure.
(d) Calculate the relative shear displacement Δx of the pin across the 25 mm gap.

2. Given Parameters & Knowns:

- Tensile load: F = 45 kN = 45,000 N | Loading mode: Double shear (resisted by two parallel planes)
- Pin: Diameter d = 16 mm = 0.016 m | Shear gap span: h = 25 mm = 0.025 m
- Material: G = 80 GPa = 8.0×10^{10} Pa, τ_{uts} = 320 MPa = 3.20×10^{8} Pa

3. Symbolic Algebraic Derivation:

1. In double shear, the applied load F is distributed across two identical cross-sectional planes of the pin:

$$A_{single} = \pi d^2 4, \quad A_{total} = 2 A_{single} = \pi d^2 2 \quad \Rightarrow \quad \boldsymbol{\tau} = F 2 A_{single} = 2 F \pi d^2$$

2. Hooke's Law for shear gives shear strain γ and lateral displacement Δx across span h:

$$\tau = G \gamma = G \Delta x h \quad \Rightarrow \quad \boldsymbol{\Delta x} = \tau h G \quad \text{(Safety Factor: } N = \tau_{uts} / \tau)$$

4. Numerical Calculation with Explicit SI Units:

- **Shear Areas:** $A_{single} = \pi(0.016)^2/4 =$ **2.0106×10^{-4} m²** (201.1 mm²) | $A_{total} =$ **4.0212×10^{-4} m²**
- **Average Shear Stress:** $\tau = 45{,}000 / (4.0212 \times 10^{-4}) = 1.1191 \times 10^{8}$ Pa = **111.91 MPa**
- **Factor of Safety:** $N = \tau_{uts} / \tau = 320 / 111.91 =$ **2.86** (adequately exceeds required structural factor $N \geq 2.5$)
- **Shear Strain & Deflection:** $\gamma = \tau / G = (1.1191 \times 10^{8}) / (8.0 \times 10^{10}) =$ **0.001399 rad** (1.40 mrad)
- **Relative Shear Displacement:** $\Delta x = \gamma h = 0.001399 \times 0.025 = 3.497 \times 10^{-5}$ m = **0.0350 mm (35.0 µm)**.

✓ **5. Physical Sanity & Boundary Check:**

- **Single vs. Double Shear:** In single shear, the stress would be 223.8 MPa, yielding a dangerously low safety factor of 1.43. Double shear halves the stress, restoring a robust safety margin.
- **Displacement Scale:** A deflection of 35 µm across a 25 mm gap represents an imperceptible elastic distortion, confirming high mechanical joint rigidity.

Worked Problem 5.3: Spherical Deep-Submergence Pressure Hull under Ocean Hydrostatic Compression

1. Problem Statement:

A deep-submergence exploration vehicle features a spherical crew compartment hull forged from aerospace titanium alloy (Ti-6Al-4V Grade 5). The vessel dives to the bottom of the Challenger Deep in the Mariana Trench at depth h = 10,900 m. Average seawater density is ρ_{sea} = 1,030 kg/m^3, and g = 9.81 m/s^2. The spherical hull has an outer radius R_0 = 1.10 m and uniform wall thickness t = 65 mm. The titanium alloy has bulk modulus B = 110 GPa, Young's modulus Y = 114 GPa, Poisson's ratio ν = 0.33, and compressive yield strength σ_y = 880 MPa.

(a) Calculate the external hydrostatic pressure P at the ocean floor.
(b) Determine the fractional volume reduction $\Delta V / V_0$ and absolute change in external volume of the sphere.
(c) Calculate the compressive membrane stress σ_{comp} induced in the spherical hull wall.
(d) Determine the safety factor against compressive yield failure.

2. Given Parameters & Knowns:

- Dive depth: h = 10,900 m | Seawater: ρ_{sea} = 1,030 kg/m^3, g = 9.81 m/s^2
- Hull: Outer radius R_0 = 1.10 m ⇒ Volume $V_0 = (4/3)\pi R_0^3 \approx 5.5753$ m^3 | Wall thickness: t = 65 mm = 0.065 m
- Material: B = 110 GPa = 1.10 × 10^{11} Pa, Y = 114 GPa, ν = 0.33, σ_y = 880 MPa = 8.80 × 10^8 Pa

3. Symbolic Algebraic Derivation:

1. Hydrostatic gauge pressure at depth h is $P = \rho_{sea}\, g\, h$. Bulk modulus relates pressure to volumetric shrinkage:

$$B = -\frac{\Delta P}{\Delta V / V_0} \quad \Rightarrow \quad \frac{\Delta V}{V_0} = -\frac{P}{B} \quad \Rightarrow \quad \Delta V = -\frac{P V_0}{B}$$

2. Hemispherical equilibrium of the spherical shell relates external pressure P to wall compressive membrane stress:

$$\Sigma F_z = 0 \quad \Rightarrow \quad P(\pi R_0^2) = \sigma_{comp}(2\pi R_0 t) \quad \Rightarrow \quad \boldsymbol{\sigma_{comp}} = \frac{P R_0}{2t} \quad (N = \sigma_y / \sigma_{comp})$$

4. Numerical Calculation with Explicit SI Units:

- **Hydrostatic Pressure:** P = 1030 × 9.81 × 10,900 = 1.1014 × 10^8 Pa = **110.14 MPa** (1,087 atmospheres!)
- **Volume Contraction:** $|\Delta V / V_0| = P / B$ = 110.14 × 10^6 / 1.10 × 10^{11} = **0.100%** | $|\Delta V|$ = 0.001001 × 5.575 = **5.58 liters**
- **Compressive Membrane Stress:** Thin-shell σ_{comp} = (110.14 × 10^6 × 1.10) / (2 × 0.065) = **931.9 MPa**. Thick-wall Lamé formula yields σ_{in} = **642.5 MPa**.
- **Factor of Safety:** $N = \sigma_y / \sigma_{actual}$ = 880 / 642.5 = **1.37** (meets hadal-depth manned submersible code 1.3–1.5).

✓ **5. Physical Sanity & Boundary Check:**

- **Ocean Shortcut Rule (Trick 4):** 10,900 m × 1.0 MPa / 100 m ≈ 109 MPa (within 1% of exact 110.14 MPa).
- **Rigidity Scale:** A 5.58 m^3 titanium sphere shrinking by only 5.58 L under 1,087 atm demonstrates extraordinary stiffness.
- **Buckling Criterion:** Spherical shells under external pressure must also be verified against elastic bifurcation buckling: $P_{cr} \approx 0.365\, Y\, (t/R)^2 \approx 143$ MPa > 110 MPa (safe).

Worked Problem 5.4: Bimetallic Constrained Structural Assembly under Thermal Shift

1. Problem Statement:

A composite structural column consists of a solid cylindrical carbon steel core surrounded by an extruded aluminum alloy (6061-T6) outer sleeve. Both components are firmly joined to rigid end plates and placed snugly between two unyielding, immovable walls at room temperature $T_0 = 20°C$. Original length is $L_0 = 0.80$ m. The cross-sectional areas and material properties are:

• Steel core: $A_s = 6.0 \times 10^{-4}$ m^2, $Y_s = 200$ GPa, $\alpha_s = 12 \times 10^{-6}$ K^{-1}

• Aluminum sleeve: $A_a = 12.0 \times 10^{-4}$ m^2, $Y_a = 70$ GPa, $\alpha_a = 24 \times 10^{-6}$ K^{-1}

The system temperature is raised uniformly to $T = 70°C$ ($\Delta T = +50°C$). Assuming no lateral buckling:
(a) Formulate the displacement compatibility condition.
(b) Determine the compressive normal stresses σ_s and σ_a developed in the steel and aluminum.
(c) Calculate the total reaction force F_{total} exerted by the rigid walls against the composite column.

2. Given Parameters & Knowns:

• Column length: $L_0 = 0.80$ m | Temperature rise: $\Delta T = 70 - 20 = +50$ K ($+50°C$)

• Steel core: $A_s = 6.0 \times 10^{-4}$ m^2, $Y_s = 200$ GPa, $\alpha_s = 12 \times 10^{-6}$ K^{-1}

• Aluminum sleeve: $A_a = 12.0 \times 10^{-4}$ m^2, $Y_a = 70$ GPa, $\alpha_a = 24 \times 10^{-6}$ K^{-1} | Boundaries: Rigid immovable walls

3. Symbolic Algebraic Derivation:

1. Rigid immovable walls enforce zero net elongation on each material independently: $\Delta L_{s,total} = 0$ and $\Delta L_{a,total} = 0$.
2. Superposition of free thermal strain and elastic mechanical strain ($\varepsilon_{total} = \alpha \Delta T + \sigma / Y = 0$) yields:

$$\boldsymbol{\sigma_s} = -\, Y_s\, \alpha_s\, \Delta T \quad \text{and} \quad \boldsymbol{\sigma_a} = -\, Y_a\, \alpha_a\, \Delta T$$

3. Internal compressive reactions and total restraining force exerted by the rigid walls:

$$F_s = |\sigma_s|\, A_s, \quad F_a = |\sigma_a|\, A_a \quad \Rightarrow \quad \mathbf{F_{total}} = F_s + F_a = (Y_s\, A_s\, \alpha_s + Y_a\, A_a\, \alpha_a)\, \Delta T$$

4. Numerical Calculation with Explicit SI Units:

• **Compressive Stresses:** $\sigma_s = -\,(200 \times 10^9)(12 \times 10^{-6})(50) = \mathbf{-120.0\ MPa}$ | $\sigma_a = -\,(70 \times 10^9)(24 \times 10^{-6})(50) = \mathbf{-84.0\ MPa}$

• **Component Reactions:** $F_s = (1.20 \times 10^8)(6.0 \times 10^{-4}) = \mathbf{72.0\ kN}$ | $F_a = (8.40 \times 10^7)(12.0 \times 10^{-4}) = \mathbf{100.8\ kN}$

• **Total Wall Reaction:** $F_{total} = 72.0 + 100.8 = \mathbf{172.8\ kN}$.

✓ **5. Physical Sanity & Boundary Check:**

• **Fast-Track Verification (Trick 5):** Steel $S_{th} = 2.40$ MPa/°C $\Rightarrow -2.4 \times 50 = -120$ MPa. Aluminum $S_{th} = 1.68$ MPa/°C $\Rightarrow -1.68 \times 50 = -84$ MPa. Exact match!

• **Yield Margin:** Steel $\sigma_{y,s} \geq 250$ MPa > 120 MPa; Aluminum $\sigma_{y,a} \geq 275$ MPa > 84 MPa (both safely elastic).

• **Length Invariance:** Whether $L_0 = 0.8$ m or 80 m, the internal stresses remain identical, underscoring the thermal stress independence theorem.

Worked Problem 5.5: Dynamic Drop-Weight Impact Loading & Elastic Energy Storage

1. Problem Statement:

An overhead warehouse crane features an emergency vertical arrestor rod made of solid high-strength aluminum alloy (6061-T6). The rod has length L_0 = 1.50 m, circular diameter d = 50 mm, Young's modulus Y = 70 GPa, and yield strength σ_y = 275 MPa. A test weight of mass m = 250 kg is dropped from a height h = 0.30 m above the collar at the bottom of the rod. Take g = 9.80 m/s^2.

(a) Determine the equivalent spring stiffness k of the rod and its static elongation Δ_{st}.

(b) Calculate the Dynamic Amplification Factor n.

(c) Determine the maximum instantaneous dynamic elongation Δ_{max}.

(d) Calculate the maximum dynamic tensile stress σ_{max}, and determine if the rod remains safely within the elastic range.

2. Given Parameters & Knowns:

• Rod: L_0 = 1.50 m, d = 50 mm ⇒ Area $A = \pi d^2 / 4 \approx 1.9635 \times 10^{-3}$ m^2 | Aluminum 6061-T6: Y = 70 GPa, σ_y = 275 MPa

• Drop mass: m = 250 kg ⇒ Weight $W = mg = 250 \times 9.80 = 2{,}450$ N | Drop height: h = 0.30 m = 300 mm | g = 9.80 m/s^2

3. Symbolic Algebraic Derivation:

1. Axial stiffness is $k = Y A / L_0$. Static deflection under gravity alone is $\Delta_{st} = mg / k = W L_0 / (A Y)$.

2. Conservation of energy during drop impact ($mg(h + \Delta_{max}) = \tfrac{1}{2} k \Delta_{max}^2$) yields the dynamic amplification factor:

$$\mathbf{n} = 1 + \sqrt{(1 + 2h\Delta_{st})} \quad \Rightarrow \quad \Delta_{max} = n\,\Delta_{st}, \quad \boldsymbol{\sigma_{max}} = n\,\sigma_{st} = n\,mgA$$

4. Numerical Calculation with Explicit SI Units:

• **Rod Axial Stiffness:** $k = Y A / L_0 = [(7.0 \times 10^{10})(1.9635 \times 10^{-3})] / 1.50 =$ **9.163×10^7 N/m** (91.63 MN/m)

• **Static Deflection & Stress:** $\Delta_{st} = mg / k = 2450 / (9.163 \times 10^7) =$ **0.02674 mm (26.74 µm)** | $\sigma_{st} = 2450 / A =$ **1.248 MPa**

• **Drop Ratio & Amplification:** $2h / \Delta_{st} = 0.60 / (2.6738 \times 10^{-5}) =$ **22,440** ⇒ **$n = 1 + \sqrt{(22{,}441)} = 150.80$**!

• **Peak Dynamic Elongation & Stress:** $\Delta_{max} = 150.80 \times 0.02674$ mm = **4.032 mm** | $\sigma_{max} = 150.80 \times 1.248$ MPa = **188.16 MPa**

• **Factor of Safety:** $N = \sigma_y / \sigma_{max} = 275 / 188.16 =$ **1.46** (remains safely within elastic limit $\sigma_{max} < \sigma_y$).

✓ **5. Physical Sanity & Boundary Check:**

• **Dynamic Force Amplification:** Static weight is 2.45 kN (~250 kg), but peak dynamic arresting force is $F_{dyn} = n W =$ **369.5 kN (~37.7 metric tons!)** — a 150-fold surge!

• **Energy Conservation:** Potential energy lost is $mg(h + \Delta) = 2450 \times 0.30403 = 744.9$ J. Stored elastic strain energy is $\tfrac{1}{2} k \Delta^2 = 0.5 \times (9.163 \times 10^7)(0.004032)^2 = 744.8$ J. Perfect match!

Part VI: Comprehensive Engineering Materials Reference Appendix

Appendix A: Master Mechanical & Elastic Properties of Engineering Materials

The table below provides representative values of density, Young's modulus, shear modulus, bulk modulus, Poisson's ratio, yield strength, ultimate tensile strength, and coefficient of thermal expansion at room temperature (20°C / 293 K). For isotropic metallic alloys, the tabulated moduli closely satisfy the theoretical relations $Y = 2G(1 + \nu)$ and $Y = 3B(1 - 2\nu)$.

Material & Specification	Density ρ (kg/m^3)	Young's Y (GPa)	Shear G (GPa)	Bulk B (GPa)	Poisson ν	Yield σ_y (MPa)	UTS σ_{uts} (MPa)	CTE α (10^{-6} K^{-1})
Structural Carbon Steel (ASTM A36)	7,850	200	78	160	0.28	250	400	12.0
High-Strength Alloy Steel (AISI 4340)	7,850	205	80	165	0.29	860	1,110	12.3
Structural Aluminum Alloy (6061-T6)	2,700	70	26	70	0.33	275	310	23.4
Aerospace Titanium (Ti-6Al-4V Gr 5)	4,430	114	44	110	0.33	880	950	8.6
Pure Copper (Electrolytic C11000)	8,940	117	44	135	0.34	70	220	16.5
Gray Cast Iron (ASTM Class 40)	7,200	110	44	95	0.26	—	290 (T) / 850 (C)	10.8
Structural Concrete (C30/37)	2,400	32	13	20	0.20	—	3 (T) / 38 (C)	10.0
Human Cortical Bone (Femur, long.)	1,900	18	6.8	14	0.30	110	130 (T) / 190 (C)	—
Vulcanized Natural Rubber	950	0.002–0.01	0.0007	2.0	0.499	—	25	220
Liquid Water (20°C, 1 atm)	1,000	0	0	2.2	0.50	0	0	207 (β_{vol})
Single Crystal Diamond (Carbon)	3,520	1,050	475	540	0.10	—	~3,000	1.1

Appendix B: SI Unit Conversions, Stress Dimensions & Mathematical Identities

Consistent units are vital in elasticity calculations to avoid catastrophic dimensional errors.

Physical Quantity	Standard SI Unit	Common Engineering Units	Conversion Equivalents
Stress & Pressure (σ, τ, P)	Pascal (Pa = N/m^2)	MPa, GPa, bar, atm, psi, ksi	1 MPa = 10^6 Pa = 1 N/mm^2 1 GPa = 10^9 Pa = 1,000 MPa = 1 kN/mm^2 1 bar = 10^5 Pa = 0.1 MPa 1 atm = 101,325 Pa ≈ 0.1013 MPa 1 psi = 6,894.76 Pa ≈ 6.895 kPa 1 ksi = 1,000 psi = 6.89476 MPa
Strain (ε, γ, ε_v)	Dimensionless (m/m)	%, microstrain (με)	1% strain = 0.01 = 10,000 με 1 με = 1 × 10^{-6} m/m
Young's / Shear / Bulk Modulus	Pascal (Pa)	GPa, MPa, Mpsi	Steel: 200 GPa ≈ 29 × 10^6 psi (29 Mpsi) Aluminum: 70 GPa ≈ 10 × 10^6 psi (10 Mpsi)
Strain Energy Density (u)	J/m^3 (Joules per cubic meter)	$N·m/m^3$, Pa, kJ/m^3	1 J/m^3 = 1 N/m^2 = 1 Pa 1 MJ/m^3 = 1 MPa
Thermal Expansion (α)	K^{-1} or $°C^{-1}$	μm/(m·°C), $°F^{-1}$	10^{-6} K^{-1} = 1 μm/(m·°C) 1 $°C^{-1}$ = (5/9) $°F^{-1}$